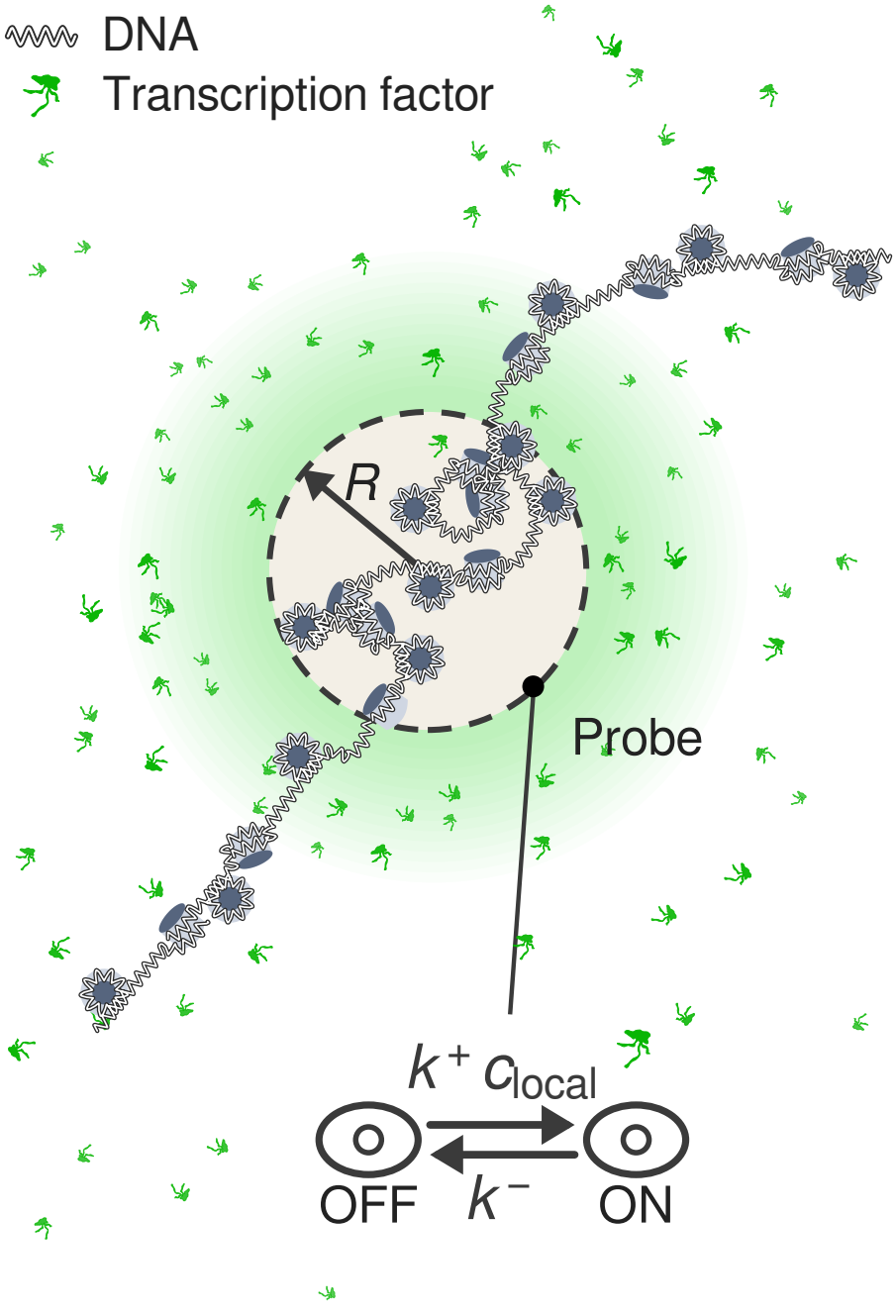


Free receptor

$$\dot{\mathcal{F}}_{\text{free}}(c) = \frac{1}{\tau_{\text{burst}}(c)}$$

Receptor at the attractive surface

$$\dot{\mathcal{F}}(c) = \underbrace{\frac{1}{\tau_{\text{burst}}(c)}}_{\text{Probe}} \underbrace{\left(\frac{d\log c_o}{d\log c}\right)^2}_{\text{Surface}}$$

$\tau_{\text{burst}}(c)$ mean cycle time of the probe

c_o TF volume fraction at the probe